ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 252 cm²	$300 - 48$
2	**(b)** 12	$b_1h_1 = b_2h_2$: $180/15$
3	**(b)** 216 cm²	Other half-diagonal $= \sqrt{15^2 - 9^2} = 12$; area $= \frac{1}{2} \times 18 \times 24$
4	**(b)** 156 m²	$(20 + 4)(15 + 4) - 20 \times 15$
5	**(d)** 150 cm²	Other leg $= \sqrt{25^2 - 15^2} = 20$; area $= \frac{1}{2} \times 15 \times 20$
6	**(d)** 50 cm²	Area $= d^2/2 = 10^2/2$
7	**(d)** 380 cm²	$450 - 70$
8	**(b)** 15	$b_1h_1 = b_2h_2$: $240/16$
9	**(d)** 480 cm²	Other half-diagonal $= \sqrt{26^2 - 10^2} = 24$; area $= \frac{1}{2} \times 20 \times 48$
10	**(b)** 324 m²	$(30 + 6)(18 + 6) - 30 \times 18$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$l + b = 40$, so $b = 14$ cm (1); area $= 26 \times 14 = 364$ cm² (1).	1+1
12	Half-diagonal$^2 = 50^2 - 14^2 = 2304$ (1); other diagonal $= 96$ cm; area $= 1344$ cm² (1).	1+1
13	Area $= d^2/2 = \frac{18^2}{2} = 162$ m² (1); side $= \sqrt{162}$ m (1).	1+1
14	$l + b = 50$, so $b = 19$ cm (1); area $= 31 \times 19 = 589$ cm² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Inner rectangle $= (40 - 4) \times (30 - 4) = 936$ m² (2); path $= 1200 - 936 = 264$ m² (1).	2+1
16	Area $= 24 \times 10 = 240$ cm² (2); height on side 16 $= 240 \div 16 = 15$ cm (1).	2+1
17	Hypotenuse $= 15$ cm (1); area $= 54$ cm² (1); altitude $= 2 \times 54/15 = 7.2$ cm (1).	1+1+1
	OR: Inner rectangle $= (50 - 6) \times (36 - 6) = 1320$ m² (2); path $= 1800 - 1320 = 480$ m² (1).	2+1
18	Area $= 30 \times 12 = 360$ cm² (2); height on side 18 $= 360 \div 18 = 20$ cm (1).	2+1
	OR: Hypotenuse $= 17$ cm (1); area $= 60$ cm² (1); altitude $= 2 \times 60/17 = 7.06$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rectangle $= 200$ cm² (1); triangle $= \frac{1}{2} \times 20 \times 8 = 80$ cm² (2); total $= 280$ cm² (1); percentage $= 28.57\%$ (1).	1+2+1+1
	OR: $h = 2A/(a + b) = 10$ cm (2); parallelogram area $= 18 \times 10 = 180$ cm² (2); triangle left $= 70$ cm² (1).	2+2+1
20	Rectangle $= 288$ cm² (1); triangle $= \frac{1}{2} \times 24 \times 10 = 120$ cm² (2); total $= 408$ cm² (1); percentage $= 29.41\%$ (1).	1+2+1+1
	OR: $h = 2A/(a + b) = 12$ cm (2); parallelogram area $= 14 \times 12 = 168$ cm² (2); triangle left $= 84$ cm² (1).	2+2+1

Mistakes to avoid:

- Using the slant side instead of the perpendicular height in the area of a parallelogram or triangle.
- Forgetting the factor $\frac{1}{2}$ in the area of a triangle, trapezium or rhombus.
- Mixing units: multiplying a length in metres by one in centimetres; convert first.
- Writing area in cm instead of cm^2, or converting m^2 to cm^2 by multiplying by 100 (it is 10 000).
- Using the sum of the non-parallel sides for a trapezium; only the **parallel** sides enter the formula.
- Adding instead of subtracting when an L-shaped region is obtained by removing a corner.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 210 cm²	Other leg $= \sqrt{29^2 - 20^2} = 21$; area $= \frac{1}{2} \times 20 \times 21$
2	**(a)** 98 cm²	Area $= d^2/2 = 14^2/2$
3	**(b)** 492 cm²	$600 - 108$
4	**(b)** 12	$b_1h_1 = b_2h_2$: $216/18$
5	**(c)** 96 cm²	Other half-diagonal $= \sqrt{10^2 - 6^2} = 8$; area $= \frac{1}{2} \times 12 \times 16$
6	**(c)** 84 m²	$(24 + 2)(16 + 2) - 24 \times 16$
7	**(b)** 180 cm²	Other leg $= \sqrt{41^2 - 9^2} = 40$; area $= \frac{1}{2} \times 9 \times 40$
8	**(c)** 128 cm²	Area $= d^2/2 = 16^2/2$
9	**(b)** 339 cm²	$384 - 45$
10	**(b)** 12	$b_1h_1 = b_2h_2$: $240/20$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Half-diagonal$^2 = 30^2 - 18^2 = 576$ (1); other diagonal $= 48$ cm; area $= 864$ cm² (1).	1+1
12	Area $= d^2/2 = \frac{22^2}{2} = 242$ m² (1); side $= \sqrt{242}$ m (1).	1+1
13	$l + b = 60$, so $b = 18$ cm (1); area $= 42 \times 18 = 756$ cm² (1).	1+1
14	Half-diagonal$^2 = 34^2 - 16^2 = 900$ (1); other diagonal $= 60$ cm; area $= 960$ cm² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Inner rectangle $= (45 - 4) \times (28 - 4) = 984$ m² (2); path $= 1260 - 984 = 276$ m² (1).	2+1
16	Area $= 36 \times 8 = 288$ cm² (2); height on side 24 $= 288 \div 24 = 12$ cm (1).	2+1
17	Hypotenuse $= 13$ cm (1); area $= 30$ cm² (1); altitude $= 2 \times 30/13 = 4.62$ cm (1).	1+1+1
	OR: Inner rectangle $= (60 - 8) \times (40 - 8) = 1664$ m² (2); path $= 2400 - 1664 = 736$ m² (1).	2+1
18	Area $= 28 \times 15 = 420$ cm² (2); height on side 20 $= 420 \div 20 = 21$ cm (1).	2+1
	OR: Hypotenuse $= 25$ cm (1); area $= 150$ cm² (1); altitude $= 2 \times 150/25 = 12$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rectangle $= 480$ cm² (1); triangle $= \frac{1}{2} \times 30 \times 12 = 180$ cm² (2); total $= 660$ cm² (1); percentage $= 27.27\%$ (1).	1+2+1+1
	OR: $h = 2A/(a + b) = 8$ cm (2); parallelogram area $= 24 \times 8 = 192$ cm² (2); triangle left $= 48$ cm² (1).	2+2+1
20	Rectangle $= 144$ cm² (1); triangle $= \frac{1}{2} \times 16 \times 6 = 48$ cm² (2); total $= 192$ cm² (1); percentage $= 25\%$ (1).	1+2+1+1
	OR: $h = 2A/(a + b) = 9$ cm (2); parallelogram area $= 20 \times 9 = 180$ cm² (2); triangle left $= 90$ cm² (1).	2+2+1

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- Mixing units: multiplying a length in metres by one in centimetres; convert first.
- Writing area in cm instead of cm^2, or converting m^2 to cm^2 by multiplying by 100 (it is 10 000).
- Using the sum of the non-parallel sides for a trapezium; only the **parallel** sides enter the formula.
- Adding instead of subtracting when an L-shaped region is obtained by removing a corner.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 600 cm²	Other half-diagonal $= \sqrt{25^2 - 15^2} = 20$; area $= \frac{1}{2} \times 30 \times 40$
2	**(a)** 276 m²	$(40 + 4)(25 + 4) - 40 \times 25$
3	**(c)** 384 cm²	Other leg $= \sqrt{40^2 - 24^2} = 32$; area $= \frac{1}{2} \times 24 \times 32$
4	**(c)** 200 cm²	Area $= d^2/2 = 20^2/2$
5	**(b)** 504 cm²	$616 - 112$
6	**(c)** 20	$b_1 h_1 = b_2 h_2$: $420/21$
7	**(b)** 384 cm²	Other half-diagonal $= \sqrt{20^2 - 12^2} = 16$; area $= \frac{1}{2} \times 24 \times 32$
8	**(a)** 366 m²	$(35 + 6)(20 + 6) - 35 \times 20$
9	**(b)** 630 cm²	Other leg $= \sqrt{53^2 - 28^2} = 45$; area $= \frac{1}{2} \times 28 \times 45$
10	**(c)** 288 cm²	Area $= d^2/2 = 24^2/2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Area $= d^2/2 = \frac{26^2}{2} = 338$ m² (1); side $= \sqrt{338}$ m (1).	1+1
12	$l + b = 60$, so $b = 25$ cm (1); area $= 35 \times 25 = 875$ cm² (1).	1+1
13	Half-diagonal$^2 = 35^2 - 21^2 = 784$ (1); other diagonal $= 56$ cm; area $= 1176$ cm² (1).	1+1
14	Area $= d^2/2 = \frac{30^2}{2} = 450$ m² (1); side $= \sqrt{450}$ m (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Inner rectangle $= (35 - 2) \times (25 - 2) = 759$ m² (2); path $= 875 - 759 = 116$ m² (1).	2+1
16	Area $= 45 \times 8 = 360$ cm² (2); height on side 30 $= 360 \div 30 = 12$ cm (1).	2+1
17	Hypotenuse $= 25$ cm (1); area $= 84$ cm² (1); altitude $= 2 \times 84/25 = 6.72$ cm (1).	1+1+1
	OR: Inner rectangle $= (48 - 4) \times (32 - 4) = 1232$ m² (2); path $= 1536 - 1232 = 304$ m² (1).	2+1
18	Area $= 42 \times 10 = 420$ cm² (2); height on side 35 $= 420 \div 35 = 12$ cm (1).	2+1
	OR: Hypotenuse $= 34$ cm (1); area $= 240$ cm² (1); altitude $= 2 \times 240/34 = 14.12$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Rectangle $= 600$ cm² (1); triangle $= \frac{1}{2} \times 40 \times 20 = 400$ cm² (2); total $= 1000$ cm² (1); percentage $= 40\%$ (1).	1+2+1+1
	OR: $h = 2A/(a + b) = 14$ cm (2); parallelogram area $= 12 \times 14 = 168$ cm² (2); triangle left $= 98$ cm² (1).	2+2+1
20	Rectangle $= 392$ cm² (1); triangle $= \frac{1}{2} \times 28 \times 10 = 140$ cm² (2); total $= 532$ cm² (1); percentage $= 26.32\%$ (1).	1+2+1+1
	OR: $h = 2A/(a + b) = 16$ cm (2); parallelogram area $= 22 \times 16 = 352$ cm² (2); triangle left $= 64$ cm² (1).	2+2+1

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